

Swapnil Pali

Senior Section Engineer (Signal)

Phone No: 8446537220 Email Id: swapnil.001pali@gmail.com Address: Nagpur, IN Date of Birth: 11-07-1993

Experienced in railway signaling design, implementation, and maintenance, with proven project management skills, strong focus on safety standards, and commitment to delivering innovative and reliable transportation solutions.

Professional Summary

- Successful design and commissioning of signalling systems for 9 stations and 5 IBS sections, resulting in enhanced track capacity and streamlined train operations.
- Led the design and implementation of Electronic Interlocking (EI) systems including RRI, PI, Automatic Signaling and **KAVACH** ensuring optimal safety and efficiency in railway operations.
- Installed one of Indian Railways' first **Direct Drive** systems at New Makardhokra Station using the **Siemens MK2 EI system's LOM card**. This will reduce use of copper cables and increase communication using OFC.
- Developed crucial signalling documentation, including Control Tables, Aspect Control Charts, Analog and Digital Circuit Designs, Cable Plans, and Bonding Plans, showcasing a strong understanding of signalling principles.
- Conducted comprehensive testing of signaling equipment and systems, including Cross Sheet Tests, SAT, IXL Testing, and Jumper Testing, to ensure system reliability and safety.
- Adept in diverse signaling technologies including Siemens and Kyosen Interlocking systems, CBTC, RRI, PI, Automatic Signaling, TWS, and ATP systems.
- Conducted training programs under Project **SAKSHAM** in Central Railway to equip technicians with the knowledge and skills required for KAVACH, Automatic Signaling, and Electronic Interlocking systems.
- Familiar with System Architecture and Design of Signaling System.
- Extensive knowledge of SIL4 safety integrity requirements and in-depth expertise in EN 50126, EN 50128 and EN 50129 railway safety standard.

Work Experience

Nov 2017 - Present

Senior Section Engineer (Signal)

Indian Railways

As Senior Section Engineer (Signal), responsible for designing, implementing, testing, and commissioning a wide range of railway signalling systems, ensuring adherence to safety regulations and operational efficiency. Played a critical role in modernizing signalling infrastructure, improving train speeds, and enhancing overall railway safety.

- Successfully installed a network of TWS Point Machines across a 70km section, enabling the operation of Vande Bharat trains at speeds of 160kmph, significantly improving travel time and passenger experience.
- Led the design, implementation, and commissioning of Electronic Interlocking systems (RRI, PI, Automatic Signalling) ensuring adherence to the highest safety standards and operational efficiency.
- Leading the installation of **Kavach, the Indian Railway Automatic Train protection (ATP) system**, in my section, ensuring successful implementation, safety compliance, and operational efficiency.

Achievements:

- I installed one of Indian Railways' first **Direct Drive** systems at New Makardhokra Station using the **Siemens MK2 EI system's LOM card**. This Central Railway project reduces copper cable use by enabling OFC-based communication.
- Installed Division First's **Kavach** System in my section.

Technical Skills

Electronic Interlocking	●●●●○
IXL Design	●●●●○
CBTC	●●●●○
Signalling Design	●●●●○
Project Management	●●●●○
Testing & Commissioning	●●●●○
Siemens Interlocking	●●●●○
RAMS	●●●○●
Kavach	●●●○●
ETCS	●●●●○
TCAS	●●●●●
PTC	●●●○●
C/C++	●●●●○
Linux	●●●●○
AutoCad	●●●●○

Soft Skills

- Teamwork
- Communication
- Problem-Solving
- Analytical Skills
- Leadership
- Adaptability
- Time Management
- Attention to Detail
- Client Management

Core Competencies

- Railway Signalling Systems Expertise
- Design & Implementation of Signalling Solutions
- Project Management & Execution
- Testing, Commissioning & Safety Assurance
- Collaboration and Teamwork

Education

- B.Tech/B.E. - Electronics/Telecommunication
- Nagpur University
- (2012 - 2016)
- Diploma In Signal Engineering

Achievements

- Enhanced safety measures in the IBS system by implementing Analog Buzzers for fire alarms, AT Tripping, and IPS failure alerts, ensuring prompt responses to critical situations.
- Divisional Level Good Performance Award 2024 for Multiple Electronic interlocking projects.
- Installed Division First's **Kavach** System in my section
- Awarded for Excellent **Leadership** and **Project Management 2025** in commissioning the Division's first **Direct Drive** system at **New Makardhokda** Station

Certifications

- **Certified Project Management Associate IPMA LEVEL D (Pursuing) (IPMA,PMA (INDIA) - 2026)**
- **Certificate Program in Gen AI for Managers** (TIMESPRO (IIM Nagpur) - 2025)
- **Kavach - Automatic Train Protection (ATP) System** (IRISET, Secunderabad (IGOTKarmyogi) - 2025)
- **Automatic Block Signalling with MSDAC** (IRISET, SECCUNDARABAD - 2025)
- **KAVACH** (IRISET, Secunderabad - 2023)
- **IRSE Certification on Power and Control** (IRSE)

Hobbies

Technology Exploration (keeping up with advancements in railway technology), Model Building (trains or other engineering projects), Travel and Photography

Languages

English, Hindi, Marathi